

The Silent Tax: Typo-Robustness Analysis (GSM8K_FIX-SPELLCHECK) DeepSeek-R1-Distill-Qwen-7B

All tables are *answered-only*: traces that hit the token cap before writing a final answer are excluded from the metrics. Configs are named `typo{rate}_real{ratio}`: *rate* = % of words corrupted, *real* = % of typos that form real words. Baseline is the clean input.

Accuracy and Flips

Table 1: Accuracy per config. *n* = questions; *n_answered* = questions that produced a final answer (not truncated at the token cap). *acc_answered* = correct / *n_answered*; *acc_strict* = correct / *n* (truncated counted as wrong); *ci_lo*/*ci_hi* = 95% bootstrap CI on *acc_strict*.

config	n	n_answered	acc_answered	acc_strict	ci_lo	ci_hi
clean	500	497	0.837	0.832	0.798	0.864
typo25_real10	500	485	0.7732	0.75	0.712	0.784
typo25_real40	500	488	0.7377	0.72	0.68	0.756
typo25_real70	500	487	0.7618	0.742	0.702	0.78
typo50_real10	500	486	0.6934	0.674	0.634	0.714
typo50_real40	500	489	0.6646	0.65	0.608	0.694
typo50_real70	500	489	0.6053	0.592	0.548	0.638
typo75_real10	500	490	0.6286	0.616	0.576	0.658
typo75_real40	500	493	0.5456	0.538	0.494	0.582
typo75_real70	500	490	0.4776	0.468	0.424	0.512

Table 2: Answered-only flips vs clean, paired by question over *n_both* (answered in both conditions). *r2w* / *w2r* = questions correct on clean but wrong on this config / wrong on clean but correct on this config; *r2w_ratio* = *r2w* / *n_both*, *w2r_ratio* = *w2r* / *n_both*; *net_flip_ratio* = *r2w_ratio* - *w2r_ratio* = (*r2w* - *w2r*) / *n_both*, the net rate at which this config worsens answers relative to clean.

config	n_both	r2w	w2r	r2w_ratio	w2r_ratio	net_flip_ratio	mcnemar_b	mcnemar_c	mcnemar_stat	mcnemar_p
typo25_real10	483	57	27	0.118	0.0559	0.0621	57	27	10.012	0.00155532
typo25_real40	486	72	19	0.1481	0.0391	0.1091	72	19	29.714	5e-08
typo25_real70	485	64	27	0.132	0.0557	0.0763	64	27	14.242	0.00016076
typo50_real10	484	92	23	0.1901	0.0475	0.1426	92	23	40.209	0.0
typo50_real40	488	103	19	0.2111	0.0389	0.1721	103	19	56.467	0.0
typo50_real70	487	129	17	0.2649	0.0349	0.23	129	17	84.39	0.0
typo75_real10	488	125	23	0.2561	0.0471	0.209	125	23	68.926	0.0
typo75_real40	490	160	17	0.3265	0.0347	0.2918	160	17	113.921	0.0
typo75_real70	487	189	16	0.3881	0.0329	0.3552	189	16	144.312	0.0

McNemar test (used in the flips table above).

- **What it is:** a paired significance test for before/after binary outcomes. It looks only at the questions that *changed* (the discordant pairs: $b = \text{right} \rightarrow \text{wrong}$, $c = \text{wrong} \rightarrow \text{right}$) and ignores those that stayed the same.
- **Null hypothesis:** typos break correct answers as often as they fix wrong ones ($b = c$, i.e. no net effect).
- **Statistic:** $(|b - c| - 1)^2 / (b + c)$ (column `mcnemar_stat`); `mcnemar_p` is its chi-square p -value with 1 degree of freedom.
- **When we reject it:** when `mcnemar_p` < 0.05 — the $\text{right} \rightarrow \text{wrong}$ vs $\text{wrong} \rightarrow \text{right}$ imbalance is too large to be chance.

Reasoning Length

Table 3: Absolute generated tokens per config (answered-only).

config	n_answered	mean_tok	median_tok	p90_tok	max_tok
clean	497	590.5	450	1098	3966
typo25_real10	485	695.2	487	1395	4096
typo25_real40	488	661.6	462.0	1429	4096
typo25_real70	487	646.4	469	1209	3972
typo50_real10	486	622.9	457.0	1179	4096
typo50_real40	489	610.5	456	1001	4096
typo50_real70	489	584.6	454	975	4096
typo75_real10	490	666.6	467.0	1236	4096
typo75_real40	493	601.9	459	946	4096
typo75_real70	490	570.3	458.5	768	4096

Table 4: Mean generated tokens for correct vs wrong answers. `wrong_over_correct` = `tok_wrong` / `tok_correct` (mean tokens of wrong answers divided by mean tokens of correct answers).

config	tok_correct	tok_wrong	wrong_over_correct
clean	574.4	673.2	1.1719
typo25_real10	660.6	813.4	1.2313
typo25_real40	649.7	695.0	1.0698
typo25_real70	629.9	699.1	1.1098
typo50_real10	636.7	591.7	0.9294
typo50_real40	593.6	644.0	1.0849
typo50_real70	595.6	567.8	0.9533
typo75_real10	614.0	755.6	1.2306
typo75_real40	581.9	626.0	1.0758
typo75_real70	529.7	607.3	1.1465

Self-doubt

The self-doubt markers are curated words grouped into two *banks*, each matched as a case-insensitive whole-word pattern in the reasoning text. Density is matches per 1000 reasoning words.

- **second_guess** — the proposal’s canonical self-correction words plus direct reversals: *wait, hmm, actually, reconsider, but wait, hold on, on second thought, let me recheck, double-check.*

- **uncertainty** — hedging / “not certain” words: *maybe, perhaps, possibly, might be, could be, not sure, I guess, presumably, or maybe / alternatively, unclear / confused.*

Example reasoning: “*Wait, maybe the total is 12... hmm, actually let me recheck*” contributes *wait, hmm, actually* (second_guess) and *maybe* (uncertainty). A separate *typo-noticing* bank (*typo, misspelled, doesn’t make sense, ...*) is treated as repair behaviour, not self-doubt.

Table 5: Doubt-marker density per config: second_guess and uncertainty, per 1k words and per trace.

config	n_answered	second_guess_per_1k	second_guess_per_trace	uncertainty_per_1k	uncertainty_per_trace	total_per_1k	total_per_trace
clean	497	7.625	1.64	3.705	0.797	11.329	2.437
typo25_real10	485	9.811	2.746	6.593	1.845	16.404	4.592
typo25_real40	488	8.918	2.328	8.706	2.273	17.623	4.6
typo25_real70	487	8.554	2.133	7.525	1.877	16.08	4.01
typo50_real10	486	8.421	1.977	9.297	2.183	17.717	4.16
typo50_real40	489	7.765	1.697	8.121	1.775	15.886	3.472
typo50_real70	489	7.044	1.466	7.506	1.562	14.55	3.029
typo75_real10	490	8.944	2.324	11.834	3.076	20.778	5.4
typo75_real40	493	6.252	1.331	10.074	2.144	16.326	3.475
typo75_real70	490	5.984	1.112	9.575	1.78	15.559	2.892

Table 6: Per-marker density (markers per 1000 reasoning words): clean vs pooled-typo. discrimination = typo minus clean (larger = more typo-responsive). All bank markers, ranked by discrimination.

marker	category	clean_per_1k	typo_per_1k	discrimination
maybe	uncertainty	1.665	4.535	2.869
or maybe/alt	uncertainty	1.038	2.075	1.037
perhaps	uncertainty	0.346	1.005	0.659
wait	second_guess	3.583	4.14	0.557
unclear/confused	uncertainty	0.29	0.558	0.268
hmm	second_guess	1.59	1.802	0.211
might be	uncertainty	0.065	0.216	0.151
not sure	uncertainty	0.037	0.12	0.083
I assume	uncertainty	0.009	0.072	0.062
actually	second_guess	0.225	0.264	0.039
not entirely	uncertainty	0.009	0.027	0.018
but wait	second_guess	0.402	0.412	0.01
reconsider	second_guess	0.0	0.0	0.0
on second thought	second_guess	0.0	0.0	0.0
possibly	uncertainty	0.009	0.004	-0.005
presumably	uncertainty	0.009	0.004	-0.005
could be	uncertainty	0.122	0.09	-0.031
I guess	uncertainty	0.103	0.071	-0.032
hold on	second_guess	0.458	0.406	-0.052
let me recheck	second_guess	0.505	0.438	-0.068
wait/actually no	second_guess	0.299	0.192	-0.107
double-check	second_guess	0.561	0.452	-0.109

Table 7: Doubt density (markers per 1000 reasoning words) split by final outcome, per category (2g = second.guess, un = uncertainty, tot = both). For a group, density = total markers in that group / total reasoning words in that group x 1000. *_correct = over correct-answer traces; *_wrong = over wrong-answer traces; *_wrong_over_correct = _wrong / _correct.

config	2g_correct	2g_wrong	2g_wrong_over_correct	un_correct	un_wrong	un_wrong_over_correct	tot_correct	tot_wrong	tot_wrong_over_correct
clean	7.608	7.699	1.012	3.233	5.839	1.806	10.841	13.537	1.249
typo25_real10	9.765	9.939	1.018	5.43	9.8	1.805	15.195	19.739	1.299
typo25_real40	9.001	8.673	0.964	8.233	10.098	1.226	17.234	18.77	1.089
typo25_real70	9.031	7.109	0.787	5.692	13.088	2.299	14.723	20.197	1.372
typo50_real10	9.268	5.945	0.642	8.927	10.378	1.163	18.194	16.324	0.897
typo50_real40	8.789	5.65	0.643	6.887	10.668	1.549	15.676	16.318	1.041
typo50_real70	7.719	5.715	0.74	8.786	4.986	0.568	16.505	10.701	0.648
typo75_real10	8.248	9.86	1.195	7.792	17.155	2.202	16.041	27.015	1.684
typo75_real40	6.718	5.689	0.847	9.05	11.315	1.25	15.767	17.003	1.078
typo75_real70	6.22	5.798	0.932	8.534	10.397	1.218	14.754	16.195	1.098

LLM judge. The proposal specifies a second, non-lexical measure for both dimensions: a separate model with a fixed, strict prompt. The judge is a different model from the one under test, called at temperature 0 on a seeded random subsample of answered traces per config. It is shown the corrupted-word list (*original* $\rightarrow$ *as shown*), the question, and the reasoning trace (head+tail window). It returns two scalar ratings and evidence: *self_doubt_score* from 0–10, where 0 means no second-guessing and 10 means pervasive uncertainty or looping; and *repair_understanding_score* from 0–5, where 0 means the question’s meaning is fully understood and 5 means the model is fundamentally lost because of the corruption. The judge also returns a one-line justification and verbatim evidence quotes; quotes are checked against the reasoning trace. The judge is a subsample measure, while the lexical tables remain the full-data result.

Repair Behaviour

Table 8: How each corrupted word was handled, per config (categories defined below the table).

config	n_corrupt_words	silent_fix_n	flagged_n	misread_n	not_used_n	silent_fix_frac	flagged_frac	misread_frac	not_used_frac
typo25_real10	3537	2395	50	24	1068	0.6771	0.0141	0.0068	0.302
typo25_real40	3574	2282	132	55	1105	0.6385	0.0369	0.0154	0.3092
typo25_real70	3548	2119	216	90	1123	0.5972	0.0609	0.0254	0.3165
typo50_real10	6926	4516	86	42	2282	0.652	0.0124	0.0061	0.3295
typo50_real40	6989	4239	218	127	2405	0.6065	0.0312	0.0182	0.3441
typo50_real70	6967	3972	353	199	2443	0.5701	0.0507	0.0286	0.3507
typo75_real10	9839	6215	126	106	3392	0.6317	0.0128	0.0108	0.3448
typo75_real40	9776	5621	363	209	3583	0.575	0.0371	0.0214	0.3665
typo75_real70	9653	5105	466	383	3699	0.5289	0.0483	0.0397	0.3832

A *corrupted word* is a word that received a typo — it differs between the clean and typo question (e.g. *sum* $\rightarrow$ *sun*). We recover these by diffing the two questions, then place each into one bucket by which form appears in the reasoning:

- **silent_fix** — only the original word (*sum*) appears: a quiet recovery.
- **flagged** — both forms appear (*sum* and *sun*): noticed and fixed.
- **misread** — only the corrupted form (*sun*) appears: read as the wrong word.
- **not_used** — neither form appears in the reasoning.

Columns *_n are counts; *_frac are the share of n_corrupt_words.

Table 9: Word handling pooled by real-word ratio: the non-word vs real-word contrast.

real_ratio	n_corrupt_words	silent_fix_frac	flagged_frac	misread_frac	not_used_frac
10	20302	0.6465	0.0129	0.0085	0.3321
40	20339	0.597	0.0351	0.0192	0.3487
70	20168	0.5551	0.0513	0.0333	0.3602

Table 10: Misread rate split by final correctness. A typo'd word is 'misread' when the reasoning uses only its corrupted form (e.g. sun instead of sum). Within each outcome group, misread rate = (typo'd words that were misread) / (all typo'd words examined), pooled over the group. misread_correct = over problems answered correctly; misread_wrong = over problems answered wrong; wrong_over_correct = misread_wrong / misread_correct.

config	misread_correct	misread_wrong	wrong_over_correct
typo25_real10	0.0066	0.0076	1.152
typo25_real40	0.0143	0.0183	1.286
typo25_real70	0.0222	0.035	1.581
typo50_real10	0.0062	0.0058	0.945
typo50_real40	0.0167	0.0207	1.236
typo50_real70	0.0283	0.0289	1.018
typo75_real10	0.0093	0.0131	1.408
typo75_real40	0.0205	0.0224	1.092
typo75_real70	0.0355	0.0431	1.216

Real-word Effect

This section isolates the effect of typo *kind* (real-word vs non-word), holding the question, corrupted positions, and rate fixed — something the vs-clean accuracy section cannot do. Three views:

- a **controlled paired** comparison of low-real vs high-real variants on the *same* questions, with a McNemar test (paired table);
- a **per-typo logistic regression** comparing the marginal harm of one real-word vs one non-word typo (logit table);
- whether wrong answers **flag the corruption** more or less as the real-word share rises (silent-failure table).

Read the signs, deltas and p -values in the tables below for this dataset's result.

Table 11: Controlled paired comparison of two TYPO variants against EACH OTHER (baseline is the low-real variant, NOT clean – so this is not the vs-clean accuracy table). Within a fixed rate the same questions are corrupted in the same positions, only the typo KIND differs. Restricted to questions answered in both variants. $\text{acc_low_real} / \text{acc_high_real}$ = accuracy on the lower / higher real-word variant; $\text{delta_acc} = \text{acc_high_real} - \text{acc_low_real}$; r2w = questions correct in the low-real variant but wrong in the high-real one; w2r = the reverse; mcnemar_p = McNemar test on that paired change. Negative delta with $\text{r2w} \neq \text{w2r}$ means real-word typos are more harmful than the non-word typos they replaced.

rate	compare	n_both	acc_low_real	acc_high_real	delta_acc	r2w	w2r	mcnemar_p
25	real10_to_real70	476	0.7794	0.7605	-0.0189	50	41	0.40167817
25	real10_to_real40	477	0.7778	0.7358	-0.0419	58	38	0.0524795
25	real40_to_real70	479	0.7411	0.7641	0.023	38	49	0.28366949
50	real10_to_real70	479	0.6931	0.6138	-0.0793	87	49	0.00151016
50	real10_to_real40	478	0.6946	0.6736	-0.0209	69	59	0.42632554
50	real40_to_real70	482	0.668	0.6079	-0.0602	66	37	0.00579924
75	real10_to_real70	482	0.6307	0.4772	-0.1535	115	41	1e-08
75	real10_to_real40	484	0.6281	0.5537	-0.0744	94	58	0.00452729
75	real40_to_real70	485	0.5505	0.4804	-0.0701	77	43	0.00259128

Table 12: Logistic regression $\text{correct} \sim \text{num_real} + \text{num_nonword}$ over all answered typo traces ($\text{num_real} / \text{num_nonword}$ = count of real-word / non-word typos in the problem). coef_^* = change in log-odds of a correct answer per one added typo of that kind (more negative = more harmful); $\text{odds_}^* = \exp(\text{coef})$; intercept is the model constant. n = number of traces fit.

n	coef_num_real	coef_num_nonword	odds_real	odds_nonword	intercept
4397	-0.07196	-0.02779	0.9306	0.9726	1.34769

Table 13: Among WRONG answers only, does the model flag the corruption, by real ratio? $\text{explicit_notice_frac}$ = fraction of wrong-answer traces whose reasoning uses strong typo-flagging language (typo / misspelled / doesn't make sense / ...); $\text{silent_frac} = 1 - \text{explicit_notice_frac}$. The proposal predicts real-word failures are quieter ($\text{explicit_notice_frac}$ should fall as real rises).

real_ratio	n_wrong	explicit_notice_frac	silent_frac
10	441	0.0907	0.9093
40	516	0.0775	0.9225
70	565	0.0761	0.9239

Lexical Grid (overview)

Table 14: All four marker families (second_guess, uncertainty, typo_noticing, repair_words) across configs.

config	second_guess	uncertainty	typo_noticing	repair_words
clean	7.625	3.705	0.851	0.73
typo25_real10	9.811	6.593	1.959	1.363
typo25_real40	8.918	8.706	3.14	2.355
typo25_real70	8.554	7.525	2.486	1.713
typo50_real10	8.421	9.297	2.795	2.156
typo50_real40	7.765	8.121	2.18	1.852
typo50_real70	7.044	7.506	2.21	1.798
typo75_real10	8.944	11.834	3.141	2.434
typo75_real40	6.252	10.074	3.374	2.316
typo75_real70	5.984	9.575	2.591	2.009

Spellcheck Recovery